

15/10/2020

Task-12

simulate Gaming concepts using pygame

Q.1

Aim: To simulate Gaming concepts using pygame

Algorithm:

- 1) Import pygame package and initialize it.
- 2) Define the window size and title.
- 3) Create a snake class which initialize the snake position, color and movement.
- 4) Create a fruit class.
- 5) Create a function to check if the snake collides.
- 6) Create a function to check if the snake collides with the window.
- 7) Create a function to update the snake position.
- 8) Create a function to update the game display and draw the snake and fruit.

Program:

#importing libraries

import pygame

import time

import random

Snake-speed = 15

#window size

window-x = 720

window-y = 480

#defining colors

black = pygame.color (0, 0, 0)

white = pygame.color (255, 255, 255)

red = pygame.color (255, 0, 0)

green = pygame.color (0, 255, 0)

blue = pygame.color (0, 0, 255)

#initialising pygame

pygame.init()

simultaneous drawing concepts using polygons

simultaneous drawing concepts using polygons

1) Input polygon points and initialize it

2) Draw the window edge and take

3) Create a state class which initializes the state

function - color and movement

outputs

1) Create a point class

2) Create a function to check if the shape collides

score=0

3) Create a function to check if the shape is in the window

the window

4) Create a function to update the state position

5) Create a function to update the score display and

draw the state and point

Program

defining variables

import pygame

import sys

import random

size = 600, 600

screen = pygame.display.set_mode(size)

pygame.display.set_caption("Simultaneous Drawing")

clock = pygame.time.Clock()

def draw_state(state):

screen.fill((0, 0, 0))

screen.fill((255, 255, 255))

screen.fill((0, 0, 0))

screen.fill((255, 255, 255))

screen.fill((0, 0, 0))

screen.fill((255, 255, 255))

screen.fill((0, 0, 0))

initialize game window.

pygame.display.set_caption('Greeks of Geeks - snakes')

game_window = pygame.display.set_mode.

FPS (frames per second)

fps = pygame.time.clock()

defining snake default position.

snake_position = [100, 50]

defining first

[90, 50],

[80, 50],

[70, 50],

]

fruit position

fruit_position = [random.randrange(1, 10,



Task - 12.2

Aim - write a python program to develop of a chess board using pygame.

Algorithm:

- 1) Import pygame and initialize it
- 2) set screen size and title
- 3) Define colors for the board and pieces.
- 4) Define a function to draw the pieces on the board by loading images for each piece and placing them on the corresponding square.
- 5) Define an initial state of the board as a list of lists containing the pieces
- 6) Draw the board and pieces on the screen

Program:

```
import pygame
# initialize pygame
pygame.init()
# set screen size and title
screen_size = (640, 640)
screen = pygame.display.set
pygame.display.set_caption('chess board')
# Define colors
black = (0, 0, 0)
white = (255, 255, 255)
brown = (153, 76, 0)
# Define function
for row in range(8):
    for col in range(8):
        square_color =
        square_rect =
        pygame.draw.rect
# Define function to draw the pieces
```


Write a Python program to develop a chess board using pygame

- 1) Import pygame and initialize it
- 2) Set screen size and title
- 3) Define colors for the board and pieces
- 4) Before a function to draw the pieces on the board
- 5) By loading images for each piece and place on the corresponding square

output

Figure 1: Chess board

	X		X	X		
X		X	X	X	X	
	X		X		X	
X		X		X		X
	X		X		X	
X		X		X	X	
	X		X		X	
X		X	X	X		

pygame.display.set_caption('Chess board')

Define colors

black = (0, 0, 0)

white = (255, 255, 255)

brown = (128, 128, 128)

Define function

for row in range(8):

for col in range(8):

square_color =

square_color =

pygame.draw.rect

Define function to draw the pieces


```

def draw_pieces(board):
    piece_images = [
    ]
    for row in range(8):
        for col in range(8):
            piece = board[row][col]
            if piece != '':
                piece_image = piece_images[piece]
                piece_rect = pygame.Rect
                screen.blit

# Define initial state of the board.
board = [
    ['r', 'n', 'b', 'q', 'k', 'b', 'n', 'r']
]

# Draw board and pieces
draw_board()
draw_pieces(board)

# Start game loop
while True:
    pygame.type.quit()
    pygame.quit()
    quit()
    pygame.display.update()

```

VEL TECH	
EX NO.	12
PERFORMANCE (5)	5
RESULT AND ANALYSIS (5)	5
VIVA VOCE (5)	5
RECORD (5)	
TOTAL (20)	
SIGN WITH DATE	15

Results

Thus, the program for the pygame is executed and verified successfully.

Completed